

CS 25000 – Computer Architecture Practice Midterm 2 (Annotated Bilingual Version)

1. An if-else construct written in assembly language contains:

A Two labels each used before they are defined

B Two labels each defined before they are used

C Two labels with one defined before use and the other used before definition

D One label use before it is defined

E None of the above answers is correct

■■ En un bloque if-else, la etiqueta del 'else' se usa antes de ser definida (porque el salto condicional va hacia adelante), mientras que la etiqueta del final del if se define antes de usarse en el salto incondicional. Por eso, una se usa antes de ser definida y la otra se define antes de usarse.

■■ In an if-else block, the 'else' label is used before being defined (since the conditional branch jumps forward), while the final label is defined before being used in the unconditional jump. Hence, one is used before defined and one is defined before used.

2. Assembly language syntax matches machine instruction format field order.

A True

B False

■■ El lenguaje ensamblador usa mnemónicos y operandos en un orden legible para humanos, que no necesariamente coincide con el orden de los campos de bits en la instrucción binaria.

■■ Assembly uses mnemonics and operands in a human-readable order, which does not necessarily match the exact bit field order in the binary instruction.

3. When a clocked register is receiving power:

A the register is not necessarily storing a bit string.

B the register waits for an input bit string before it outputs a bit string.

C the register waits for a clock input before it outputs a bit string.

D All of the above answers are correct.

E All of the above answers are incorrect.

■■ Un registro sincronizado con reloj mantiene su valor actual y solo actualiza su salida cuando llega una señal de reloj. No cambia continuamente mientras tenga energía.

■■ A clocked register holds its value and only updates when a clock edge arrives. It doesn't continuously change while powered.